Same score, different source: making negativity comparable across outlets, platforms and donated feeds

Governing question: how much negativity, and what kind, do Danish news outlets publish, do platforms surface, and do individual feeds actually contain — and can those three quantities be placed on a scale that makes the comparison mean something?

Research proposal for the PhD position in SP1, Navigating Negativity: What is a Healthy News Diet? · Centre for Journalism, University of Southern Denmark · Asad Ikram · September 2026

PROBLEM

Denmark now reports one of the sharpest rises in news avoidance in Europe, and the project asks what a healthy news diet looks like when the answer depends on how much negativity a diet contains. SP1's own question — how the amount and character of negativity *differ across* news outlets, social media platforms and individual news diets — contains a measurement assumption that the field's own evidence does not support: that one negativity instrument behaves the same way on an outlet's article text, on a platform post, and on the mixed, truncated, informal text of a donated feed. If it does not, the differences SP1 reports are partly differences between the sources and partly differences in how the instrument reads them, and there is currently no way to tell which. That matters beyond SP1, because SP3's exposure variable is built from SP1's numbers: Scharkow and Bachl (2016) showed that content-analysis error of exactly this kind attenuates linkage estimates and can manufacture minimal effects. This proposal takes that precondition as its subject.

CONSTRUCT

Negativity is not one quantity. Lengauer, Esser and Berganza (2012), whose phrase "amount and character" the project uses, distinguish negativity from confrontation, frame-related from actor-related negativity, and non-directional from directional tone; the DOCA conventions separate negative tone toward actors, overall tonality, pessimistic outlook, conflict-centredness and attributions of incapability or misconduct. Soroka, Young and Balmas (2015) show valence and the discrete emotions of fear and anger are empirically distinct. "Amount" and "character" are therefore separate estimands and a single scalar sentiment score cannot carry both. I follow the unit of analysis used in this department's own comparative work, coding tone at the level of the actor in the story rather than the document (Hopmann, de Vreese and Albæk 2011). I also treat the negativity bias as an open empirical question rather than a premise, following Andersen and Shehata's (2025) preregistered finding of a preference for positive news.

THE GAP THIS ADDRESSES

That automated negativity measures lose accuracy when moved between domains, genres and languages is this field's established premise (van Atteveldt, van der Velden and Boukes 2021; Boukes et al. 2020; Chan et al. 2021), and that imperfect machine labels bias downstream estimates has been demonstrated within a single estimation setting (Scharkow and Bachl 2016; Egami et al. 2023). To my knowledge, no study has yet treated this as a measurement-equivalence problem: whether one negativity instrument is equivalent across outlet text, platform posts and individual donated feeds, and how those scores can be linked onto a common scale — which is the precondition SP1's core comparison silently assumes. The review by Andersen, Toff and Ytre-Arne (2024) names a related blind spot, that we know little about the role of news media and intermediaries in avoidance; measuring negativity at all three levels is a measurement of precisely that contribution.

WORK PACKAGES

WP1. A Danish negativity instrument, multidimensional and validated per source. A codebook built from Lengauer et al.'s axes and the DOCA variables, coded at actor-in-story level, with valence separated from fear and anger. A gold standard annotated by Danish-native coders, two per item with an iterated coding list, and Krippendorff's alpha reported per construct *and per source*. A multilingual encoder is fine-tuned and benchmarked honestly against AFINN, SENTIDA, the DaNLP tone model and an LLM first pass, since on Danish data the lexical tools collapse on exactly the formal-to-informal transition this project depends on (AFINN 0.68 on Europarl to 0.48 on Twitter) while transformer tone models hold better (0.79 to 0.73). No published Danish sentiment resource is validated on Danish news, and SENTIDA's own domain validation rests on 45 short texts. *Limitation:* a gold standard of the size three years permits will bound the ceiling on rare constructs such as attributions of misconduct.

WP2. The donation hub as a measuring instrument. The hubs are the project's methodological innovation, and I treat them as an acquisition instrument whose error properties must be known, not as fieldwork. Concretely: a schema-drift detection and repair layer over download packages, with golden fixtures per platform and per locale, versioned parsers stamped onto every record, and canary donations from project-owned accounts — because package structure changes with the account's language setting, which is decisive in Denmark where donors' phones split between Danish and English. A content-resolution layer, because donated entries are identifiers and links rather than text and a classifier cannot read them; this is the step Hase et al. (2024) name as the blocker that makes content analysis of donated feeds unfeasible. Selective redaction with a public-entity allowlist, since blanket handle-redaction would erase the outlet identity on which the whole comparison depends. On-device extraction with two-stage consent, building on Port. *Limitation:* resolution will fail for deleted and restricted items, biasing the recovered feed toward content that is still live, which is a survivorship bias correlated with the outcome; I will measure and report resolution rate by platform, item age and outlet rather than assume it away.

WP3. Equivalence and linking. Test whether the WP1 instrument is measurement-equivalent across the three text types; model document length as an explicit facet, because Chan et al. (2021) found the dominant latent factor across dictionary scores correlates almost perfectly with article length, and length is exactly what separates an article from a post from a caption; test differential item functioning by outlet, platform and format; and use score linking so a feed-level score sits on the same scale as an outlet-level one. Classifier error is then propagated into the aggregates SP2 and SP3 consume rather than discarded. *Limitation:* linking requires anchor content appearing in more than one context, and the design's first task is to establish that enough exists.

WP4. The substantive payoff. Do individual Danish feeds actually contain more negativity than the outlets feeding them? Neither survey self-report nor a simulated feed can settle this; donated feeds can. The output is a person-day exposure measure with a stated error model, shaped for joining to SP3's panel in the way Hopmann, Vliegenthart, de Vreese and Albæk (2010) shaped daily content data for rolling cross-section survey data. *Limitation:* donated feeds are a self-selected sample of a self-selected platform population.

POPULATION, FRAME AND SAMPLE

The outlet frame is drawn from the Danish national and regional press via a media archive with random-day sampling across the study period, extended to alternative outlets so the comparison is not confined to legacy brands. Donated feeds are treated as a non-probability sample and benchmarked against SP3's panel on observable demographics. Expected yield is planned from published figures rather than optimism: Keusch et al. (2024) report 79 per cent willing and 48 per cent of those succeeding, with most failures technical, which is the argument for an assisted hub in the first place. Every funnel step is logged, so the hub produces data about itself and a methods paper about assisted physical donation, which to my knowledge has not been evaluated.

MEASUREMENT QUALITY, ETHICS AND LANGUAGE

Reliability is reported before performance: alpha per construct per source alongside macro-F1, and an LLM first pass validated against the Danish gold standard rather than trusted, since SDU's own Danish benchmark work finds frontier models weak on Denmark-specific material. Language is treated as a third axis of non-transfer alongside domain and genre, so an instrument built for English social media applied to Danish news copy is a triple shift, and language enters the invariance test as a facet rather than being handled once at the start. I do not speak Danish: the annotation, the codebook and the qualitative reading of Danish text are done by native speakers in the team and by recruited coders, my contribution is the instrument, the infrastructure and the analysis, and I would begin Danish tuition on arrival. Ethics and data protection are designed in, not appended: a data protection impact assessment before any donation, two-stage consent with the donor inspecting the extracted preview before releasing it, on-device extraction so the raw package never leaves the phone, selective redaction validated against a re-identification assessment, explicit handling of third parties and of donors who were minors when the account was used, and storage on university infrastructure under SDU governance.

THREE-YEAR PLAN AND CONTRIBUTION

Year 1: codebook, gold standard and WP1 instrument; parser and resolution layer built against public fixtures; hub protocol piloted on a small assisted sample. Year 2: hub fielded; WP3 equivalence and linking study. Year 3: WP4 linkage, the handover to SP3, and thesis. If hub yield underperforms, WP1 and WP3 stand on outlet and platform text alone and the feed layer becomes a smaller validation study rather than the spine. The contribution is three things: the first Danish news negativity instrument validated per source, an evaluation of assisted physical data donation as a method, and a negativity estimate solid enough to underwrite the public advice on news diets that this project exists to give.

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